Chapter 9 Infinite Sequences and Series

Section 9.1 Infinite Sequences

A sequence is a list of numbers $a_1, a_2, a_3, \cdots, a_n, \cdots$ in a given order. Each of a_1, a_2, a_3, and so on are the **terms** of the sequence. a_1 and a_n are the 1st term and n-th term of the sequence, respectively. The integer n is called the **index** of a_n and indicates where a_n occurs in the list.
More precisely, an **infinite sequence** of numbers is a function whose domain is the set of positive integers. For example, the function $f(n) = 2n$, $n \in \mathbb{N}$, associated with the sequence

$$2, 4, 6, 8, 10, 12, \cdots, 2n, \cdots.$$

Sequences can be described by writing rules (or formulas) that specify their terms, such ss

$$a_n = \sqrt{n}\,, b_n = \frac{(-1)^{n+1}}{n}\,, c_n = \frac{n-1}{n}\,, d_n = (-1)^{n+1}\,,$$

or by listing terms:

$$\{a_n\} = \{\sqrt{1}, \sqrt{2}, \sqrt{3}, \cdots, \sqrt{n}, \cdots\}$$

$$\{b_n\} = \left\{1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \cdots, \frac{(-1)^{n+1}}{n}, \cdots\right\}$$

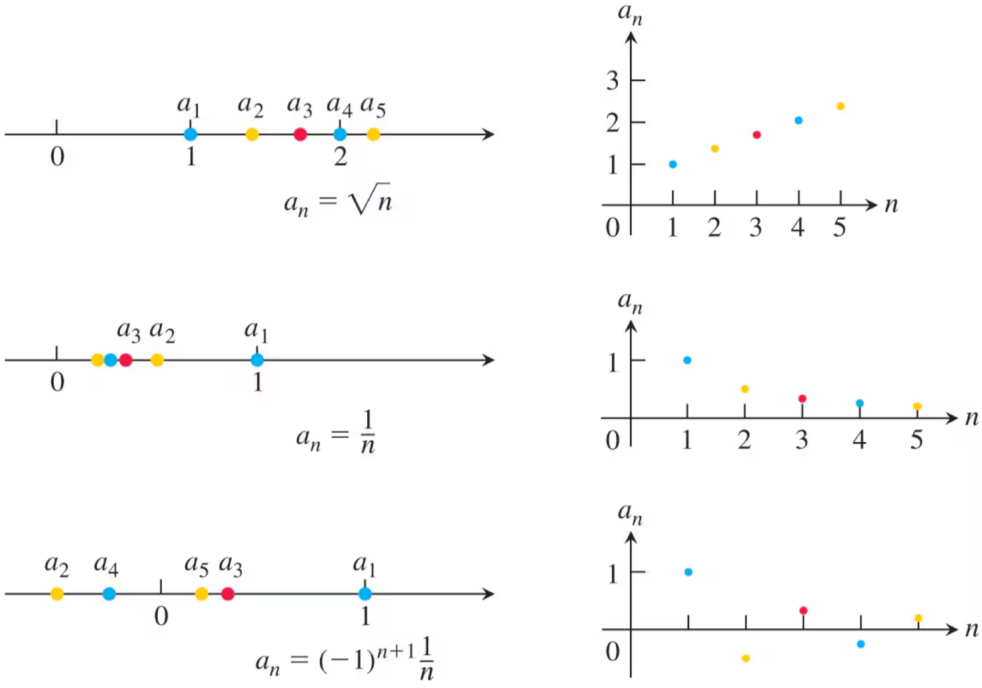
$$\{c_n\} = \left\{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \cdots, \frac{n-1}{n}, \cdots\right\}$$

$$\{d_n\} = \{1, -1, 1, -1, \cdots, (-1)^{n+1}, \cdots\}$$

A sequence also can be describes using its rules, as with $\{a_n\} = \{\sqrt{n}\}_{n=1}^{\infty}$

Notation:
Let a_n be the formula that describes a sequence. Then $\{a_n\}_{n=1}^{\infty}$ denote an infinite sequence while $\{a_n\}_{n=1}^{k}$, k is a positive integer, denote a finite sequence with k terms.

The following figure shows two ways to represent sequences graphically. The first marks the first few points from $a_1, a_2, a_3, \cdots, a_n, \cdots$ on the real axis. The second method shows the graph of the function defining the sequence. The function is defined only on integer inputs.

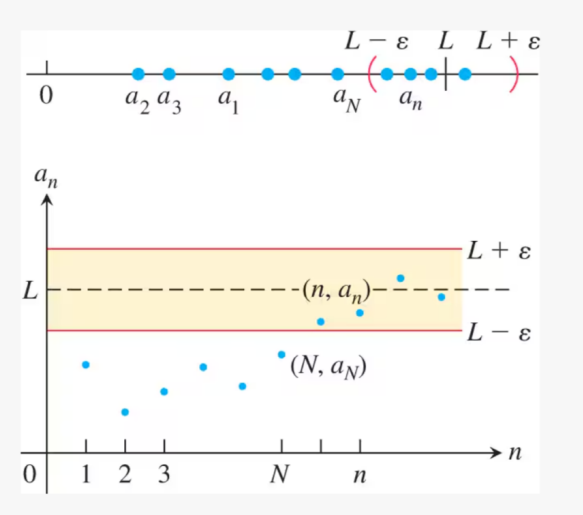


Convergence and divergence

Definition 1:
The sequence $\{a_n\}$ **converges** to the number L if for every positive number ε there corresponds an integer N such that

$$| a_n - L | < \varepsilon \quad \text{whenever} \quad n > \text{N}.$$

If no such number L exists, we say that $\{a_n\}$ **diverges**.
If an converges to L, we write $\lim_{n\to\infty} a_n = L$, or simply $a_n \to L$, and call L the **limit** of the sequence.



In the representation of a sequence as points in the plane, $a_n \to L$ if $y = L$ is a horizontal asymptote of the sequence of points $\{(n, a_n)\}$. In this figure, all the a_n's after a_N lie within ε of L.

Definition 2:
The sequence an **diverges to infinity** if for every number M there is an integer N such that for all n larger than N, $a_n > M$. If this condition holds, then we write

$$\lim_{n\to\infty} a_n = \infty \quad \text{or} \quad a_n \to \infty .$$

Similarly, if for every number m there is an integer N such that for all $n > N$ we have $a_n < m$, then we say an **diverges to negative infinity** and write

$$\lim_{n\to\infty} a_n = -\infty \quad \text{or} \quad a_n \to -\infty .$$

A sequence may diverge without diverging to infinity or negative infinity, as in Example 1. The sequences $\{1, -2, 3, -4, 5, -6, 7, -8, \cdots\}$ and $\{1, 0, 2, 0, 3, 0, \cdots\}$ are also sequences that diverge but do not diverge to infinity or negative infinity.

Remark:

1. $\lim_{n\to\infty} a_n = L$ (a finite number) $\Rightarrow \{a_n\}_{n=1}^{\infty}$ converges to L.

2. $\lim_{n\to\infty} a_n$ does not exists $\Rightarrow \{a_n\}_{n=1}^{\infty}$ diverges.

3. $\{a_n\}_{n=1}^{\infty}$ diverges to infinity if $\lim_{n\to\infty} a_n = \infty$ and it diverges to negative infinity is $\lim_{n\to\infty} a_n = -\infty$.

   A sequence may diverge without diverging to infinity or negative infinity.

Example 1:
Which of the following sequences converge and which diverge?

(i) $\{2, \sqrt{5}, \sqrt{6}, \sqrt{7}, 2\sqrt{2}, \cdots\}$ (ii) $\{-1, 1, -1, 1, \cdots\}$ (iii) $\{\sin n\}_{n=1}^{\infty}$

Solution:

(i) $a_n = \sqrt{n+3}$, $n = 1, 2, 3, \cdots$.

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \sqrt{n+3} = \infty$. $\therefore$ $\{\sqrt{n+3}\}_{n=1}^{\infty}$ diverges to ∞.

(ii) $a_n = (-1)^{n+1}$, $n = 1, 2, 3, \cdots$.

As $n \to \infty$, $(-1)^{n+1}$ alternates between -1 and 1.

$\therefore$ $\lim_{n\to\infty}(-1)^{n+1}$ does not exist. $\therefore$ $\{(-1)^{n+1}\}_{n=1}^{\infty}$ diverges.

(iii) As $n \to \infty$, the values of sin n cycle from -1 to 1.

$\therefore$ $\lim_{n\to\infty} \sin n$ does not exist. $\therefore$ $\{\sin n\}_{n=1}^{\infty}$ diverges.

Calculating Limits of the Sequences

Theorem 1
Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers, and let A and B be real numbers. The following rules hold if $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$.

1. *Sum Rule:*	$\lim_{n\to\infty}(a_n + b_n) = A + B$
2. *Difference Rule:*	$\lim_{n\to\infty}(a_n - b_n) = A - B$
3. *Constant Multiple Rule:*	$\lim_{n\to\infty}(k \cdot b_n) = k \cdot B$ (any number k)
4. *Product Rule:*	$\lim_{n\to\infty}(a_n \cdot b_n) = A \cdot B$
5. *Quotient Rule:*	$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}$ if $B \neq 0$

Example 2:

Since $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ and $\{3\}_{n=1}^{\infty}$ converge to 0 and 3, respectively, by Theorem 1, $\left\{\frac{1}{n}+3\right\}_{n=1}^{\infty}$, $\left\{\frac{3}{n}\right\}_{n=1}^{\infty}$ and $\left\{-\frac{1}{n}\right\}_{n=1}^{\infty}$ converge to 3, 0 and 0, respectively.

Cautions:

1. $\{a_n + b_n\}$ converges **<u>does not imply</u>** $\{a_n\}$ and $\{b_n\}$ converge.

2. **$\{a_n\}$ diverges $\Rightarrow$ $\{ca_n\}$, $c \neq 0$, diverges.** (Example 4)

Example 3:
$\{a_n\} = \{1, 2, 3, \cdots, \cdots\}$ and $\{b_n\} = \{-1, -2, -3, \cdots, -n, \cdots\}$ both are divergent sequences but

$\{a_n + b_n\} = \{0, 0, 0, \cdots\}$ converges to 0.

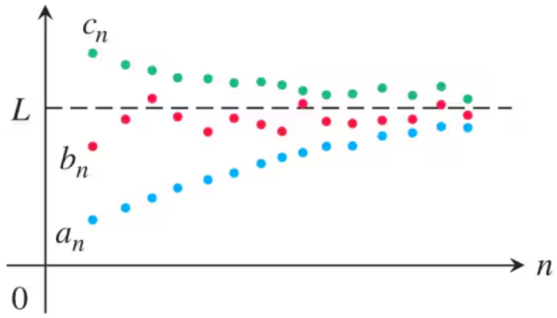
Example 4:
Show that every nonzero multiple of a divergent sequence diverges.

Solution:
Suppose, to the contrary, that $\{ca_n\}$ converges for some number $c \neq 0$. Then, by the Constant Multiple Rule in Theorem 1, $\{a_n\} = \left\{\frac{1}{c}(ca_n)\right\}$ converges. This contradicts $\{a_n\}$ is a divergent sequence. Hence, $\{ca_n\}$, $c \neq 0$, diverges when $\{a_n\}$ diverges.

Example 5:
Which of the following sequences converge and which diverge?

(i) $\left\{\frac{n+1}{n}\right\}_{n=1}^{\infty}$ (ii) $\left\{\frac{6-7n^6}{n^6+5}\right\}_{n=1}^{\infty}$

Theorem 2 — The Sandwich Theorem for Sequences
Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers. If $a_n \le b_n \le c_n$ holds for all n beyond some index N, and if $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$ also.



The terms of sequence b_n are sandwiched between those of a_n and c_n, forcing them to the same common limit L.

Example 6:
Since $1/n \to 0$. then

(a) $\frac{\cos n}{n} \to 0$	because	$-\frac{1}{n} \le \frac{\cos n}{n} \le \frac{1}{n}$;
(b) $\frac{1}{2^n} \to 0$	because	$0 \le \frac{1}{2^n} \le \frac{1}{n}$;
(c) $(-1)^n \frac{1}{n} \to 0$	because	$-\frac{1}{n} \le (-1)^n \frac{1}{n} \le \frac{1}{n}$.
(d) If $\lvert a_n \rvert \to 0$, then $a_n \to 0$	because	$-\lvert a_n \rvert \le a_n \le \lvert a_n \rvert$.

Theorem 3 — The Continuous Function Theorem for Sequences
Let $\{a_n\}$ be a sequence of real numbers. If $a_n \to L$ and if f is a function that is continuous at L and defined at all a_n then $f(a_n) \to f(L)$.

Example 7:

Show that $\sqrt{\frac{n+1}{n}} \to 1$ as $n \to \infty$.

Solution

We know that $\frac{n+1}{n} \to 1$ as $n \to \infty$. Taking $f(x) = x$ and $L = 1$ in Theorem 3 gives $\sqrt{\frac{n+1}{n}} \to 1$ as $n \to \infty$.

Using L'Hôpital's Rule

Theorem 4:
Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that an is a sequence of real numbers such that $a_n = f(n)$ for $n \geq n_0$. Then $\lim_{x\to\infty} f(x) = L \implies \lim_{n\to\infty} a_n = L$.

Example 8:
Which of the following sequences converge and which diverge?

(i) $\left\{\frac{(-1)^n}{n^2}\right\}_{n=1}^{\infty}$ (ii) $\left\{(-1)^n(1+\frac{1}{n^2})\right\}_{n=1}^{\infty}$ (iii) $\left\{ne^{-n}\right\}_{n=1}^{\infty}$

Solution:

(i) $a_n = \frac{(-1)^n}{n^2} = \begin{cases} -\frac{1}{n^2}, & n \text{ is odd} \\ \frac{1}{n^2}, & n \text{ is even} \end{cases}$.

$\therefore \quad \frac{-1}{n^2} \leq \frac{(-1)^n}{n^2} \leq \frac{1}{n^2} \quad \forall \ n = 1, 2, 3, \ldots.$

$\lim_{n\to\infty} \frac{1}{n^2} = 0 = \lim_{n\to\infty}\left(-\frac{1}{n^2}\right)$. By the sandwich theorem, $\lim_{n\to\infty} \frac{(-1)^n}{n^2} = 0$.

$\therefore \quad \left\{\frac{(-1)^n}{n^2}\right\}_{n=1}^{\infty}$ converges to 0.

(ii) $a_n = (-1)^n\left(1+\frac{1}{n^2}\right) = \begin{cases} -\left(1+\frac{1}{n^2}\right), & n \text{ is odd} \\ 1+\frac{1}{n^2}, & n \text{ is even} \end{cases}$.

As $n \to \infty$, a_n alternates between1 and –1. Hence, $\lim_{n\to\infty} a_n$ does not exist.

$\therefore \ \left\{(-1)^n\left(1+\frac{1}{n^2}\right)\right\}_{n=1}^{\infty}$ diverges.

(iii) Let $f(x)=\frac{x}{e^x}$, $x\geq 1$. Then the function agrees with the given sequence at positive integers.

By Theorem 4, $\lim_{n\to\infty} ne^{-n} = \lim_{n\to\infty}\frac{n}{e^n} = \lim_{x\to\infty}\frac{x}{e^x} = \lim_{x\to\infty}\frac{1}{e^x} = 0.$

$\therefore$ $\{ne^{-n}\}_{n=1}^{\infty}$ converges to 0.

$\lim_{n\to\infty}\frac{n}{e^n}$ is an indeterminate form of type $\frac{\infty}{\infty}$. l'Hopital's rule is indicated.
Howerever, **CANNOT** apply the rule **DIRECTLY** because both $f(n)=n$ and $g(n)=e^n$ are defined for **positive integer *n* only,** and hence are **NOT DIFFERENTAIBLE.**
In this case, we need to **extend the domain of these function** and then apply Theorem 4.

Example 9:

Does the sequence whose nth term is $a_n=\left(\frac{n+1}{n-1}\right)^n$, $n\geq 2$, converge? If so, find $\lim_{n\to\infty} a_n$.

Solution:

The function $f(x)=\left(\frac{x+1}{x-1}\right)^x$ is defined for all real numbers $x\geq 2$ and agrees with a_n at all integers $n\geq 2$. If we can show that $\lim_{x\to\infty}\left(\frac{x+1}{x-1}\right)^x = L$, then by Theorem 4, $\lim_{n\to\infty}\left(\frac{n+1}{n-1}\right)^n = L$.

The limit $\lim_{x\to\infty} f(x)$ leads to the indeterminate form 1^∞. To evaluate this limit, we apply l' Hôpital's Rule after taking the natural logarithm of $f(x)$:

$$\ln f(x)=\ln\left(\frac{x+1}{x-1}\right)^x = x\ln\left(\frac{x+1}{x-1}\right).$$

$$\begin{aligned}
\lim_{x\to\infty}\ln f\left(x\right) &= \lim_{x\to\infty} x\ln\left(\tfrac{x+1}{x-1}\right) && \infty\cdot 0 \text{ form}\\
&= \lim_{x\to\infty}\frac{\ln\left(\frac{x+1}{x-1}\right)}{1/x} && \tfrac{0}{0} \text{ form}\\
&= \lim_{x\to\infty}\frac{-2/(x^2-1)}{-1/x^2} && \text{Apply I'Hôpital's Rule.}\\
&= \lim_{x\to\infty}\frac{2x^2}{x^2-1} = 2. && \text{Simplify and evaluate.}
\end{aligned}$$

Therefore, $\lim_{x\to\infty}\left(\frac{x+1}{x-1}\right)^x = \lim_{x\to\infty} f\left(x\right) = \lim_{x\to\infty} e^{\ln f(x)} = e^2$. Applying Theorem 4, we conclude that the sequence $\{a_n\}$ also converges to e^2.

Commonly Occurring Limits (Some useful Limits)

The next theorem gives some limits that arise frequently.

THEOREM 5 The following six sequences converge to the limits listed below:

1. $\lim_{n\to\infty} \frac{\ln n}{n} = 0$

2. $\lim_{n\to\infty} \sqrt[n]{n} = 1$

3. $\lim_{n\to\infty} x^{1/n} = 1 \quad (x > 0)$

4. $\lim_{n\to\infty} x^n = 0 \quad (|x| < 1)$

5. $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x \quad$ (any x)

6. $\lim_{n\to\infty} \frac{x^n}{n!} = 0 \quad$ (any x)

In Formulas (3) through (6), x remains fixed as $n \to \infty$.

x is a fixed number (constant)

Proof:

1. Let $f(x) = \frac{\ln x}{x}$. Then the function agrees with the given sequence at positive integers.

$$\lim_{n\to\infty} \frac{\ln n}{n} = \lim_{x\to\infty} \frac{\ln x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = 0..$$

2. Let $f(x) = \sqrt[x]{x}$, $x \geq 1$. Then the function agrees with the given sequence at positive integers.

$$\ln f(x) = \ln(x)^{1/x} = \frac{1}{x}\ln x$$

$$\Rightarrow \lim_{x\to\infty} \ln f(x) = \lim_{x\to\infty} \frac{\ln x}{x}$$

$$\Rightarrow \ln \lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{\frac{1}{x}}{1} \quad \text{(by l'Hopital's Rule)} \quad \Rightarrow \quad \ln \lim_{x\to\infty} f(x) = \lim_{x\to\infty} 0 = 0$$

$$\Rightarrow \lim_{x\to\infty} f(x) = e^0 = 1 \text{ . By Theorem 4, } \lim_{n\to\infty} \sqrt[n]{n} = \lim_{x\to\infty} \sqrt[x]{x} = 1 .$$

3 .

$$\lim_{n\to\infty} \ln x^{1/n} = \lim_{n\to\infty} \frac{\ln x}{n} \quad \text{, where } x > 0 \text{ is fixed}$$

$$\Rightarrow \ln \lim_{n\to\infty} x^{1/n} = (\ln x) \lim_{n\to\infty} \frac{1}{n} = 0$$

$$\Rightarrow \lim_{n\to\infty} x^{1/n} = e^0 = 1 .$$

Example 10: These are examples of the limits in Theorem 5.

(a) $\dfrac{\ln(n^2)}{n} = \dfrac{2\ln n}{n} \to 2 \cdot 0 = 0$ Formula 1

(b) $\sqrt[n]{n^2} = n^{2/n} = (n^{1/n})^2 \to (1)^2 = 1$ Formula 2

(c) $\sqrt[n]{3n} = 3^{1/n}(n^{1/n}) \to 1 \cdot 1 = 1$ Formula 3 with $x = 3$ and Formula 2

(d) $\left(-\dfrac{1}{2}\right)^n \to 0$ Formula 4 with $x = -\dfrac{1}{2}$

(e) $\left(\dfrac{n-2}{n}\right)^n = \left(1 + \dfrac{-2}{n}\right)^n \to e^{-2}$ Formula 5 with $x = -2$

(f) $\dfrac{100^n}{n!} \to 0$ Formula 6 with $x = 100$ ■

Recursive Definitions

Sequences are often defined **recursively** by giving

1. The value(s) of the initial term or terms, and
2. a rule called a **recursion formula** for calculating any later term from terms that precede it.

Example 11:

(1) The statements $a_1 = 1$ and $a_n = a_{n-1} + 1$ for $n > 1$, define the sequence 1, 2 , 3, ⋯ , n, ⋯ of positive integers. With $a_1 = 1$, we have $a_2 = a_1 + 1 = 2$, $a_3 = a_2 + 1$, and so on.

(2) The statements $a_1 = 1$ and $a_n = na_{n-1}$ for $n > 1$ define the sequence 1, 2, 6, 24, ⋯ , $n!$, ⋯ of factorials. With $a_1 = 1$, we have $a_2 = 2a_1 = 2$, $a_3 = 3a_2 = 6$, $a_4 = 4a_3 = 24$, and so on.

(3) The statements $a_1 = 1$, $a_2 = 1$ and $a_{n+1} = a_n + a_{n-1}$ for $n > 2$, define the sequence 1, 1, 2, 3, 5, ⋯ of **Fibonacci numbers**. With $a_1 = 1$ and $a_2 = 1$, we have $a_3 = 1 + 1 = 2$, $a_4 = 2 + 1 = 3$, $a_5 = 3 + 2 = 5$, and so on.

Bounded Monotonic Sequences

Definition 3:

A sequence $\{a_n\}$ is **bounded from above** if there exists a number M such that $a_n \leq M$ for all n. The number M is an **upper bound** for $\{a_n\}$. If M is an upper bound for $\{a_n\}$ but no number less than M is an upper bound for $\{a_n\}$, then M is the **least upper bound** for $\{a_n\}$.

A sequence $\{a_n\}$ is **bounded from below** if there exists a number m such that $a_n \geq m$ for all n. The number m is a **lower bound** for $\{a_n\}$. If m is a lower bound for $\{a_n\}$ but no number greater than m is a lower bound for $\{a_n\}$, then m is the **greatest lower bound** for $\{a_n\}$.

If $\{a_n\}$ is bounded from above and below, then $\{a_n\}$ is **bounded**. If $\{a_n\}$ is not bounded, then we say that $\{a_n\}$ is an **unbounded** sequence.

Example 12:

(1) The sequence 1, 2 , 3, ⋯ , n, ⋯ has no upper bound because it eventually surpasses every number M. However, it is bounded below by every real number less than or equal to 1. The number $m = 1$ is the greatest lower bound of the sequence.

(2) The sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \cdots, \frac{n}{n+1}, \cdots$. is bounded above by every real number greater than or equal to 1. The upper bound $M = 1$ is the least upper bound. The sequence is also bounded below by every number less than or equal to 12, which is its greatest lower bound.

Definition4:
A sequence $\{a_n\}$ is **nondecreasing** if $a_n \leq a_{n+1} + 1$ for all n. That is, $a_1 \leq a_2 \leq a_3 \leq \cdots$. The sequence is **nonincreasing** if $a_n \geq a_{n+1} + 1$ for all n. The sequence {an} is **monotonic** if it is either nondecreasing or nonincreasing.

Example 13:

(1) The sequence 1, 2 , 3, ⋯ , n, ⋯ is nondecreasing.

(2) The sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \cdots, \frac{n}{n+1}, \cdots$. is nondecreasing.

(3) The sequence $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \cdots, \frac{1}{2^n}, \cdots$ is nonincreasing.

(4) The constant sequence 3, 3, 3, ⋯ , 3, ⋯ is both nondecreasing and nonincreasing.

(5) The sequence 1, −1, 1, −1, 1, −1, ⋯ is not monotonic.

A sequence that is bounded from above always has a least upper bound. Likewise, a sequence bounded from below always has a greatest lower bound.

Theorem 6—The Monotonic Sequence Theorem
If a sequence an is both bounded and monotonic, then the sequence converges.